

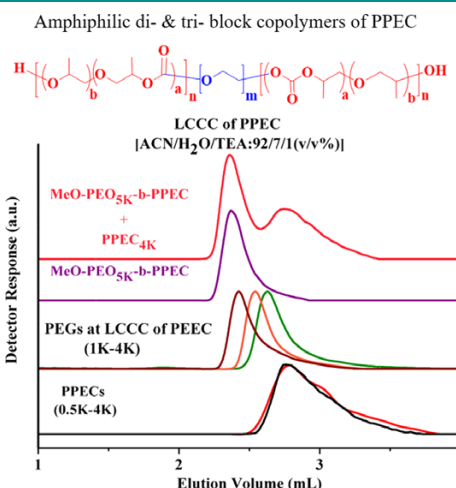
Poly(propylene ether carbonate)-Based Di- and Tri-Block Copolymers: Synthesis and Chromatographic Characterization

Rubina Abdul-Karim
Syed Ghulam Musharraf
Muhammad Imran Malik*

H.E.J. Research Institute of Chemistry, International Centre for Chemical and Biological Sciences (ICCBS),
University of Karachi, Karachi 75270, Pakistan

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Abstract: In this study, ring-opening polymerization (ROP) of propylene carbonate (PC) is conducted for synthesis of novel amphiphilic di- and tri-block copolymers containing poly(ethylene oxide) as hydrophilic with poly(propylene ether-carbonate) (PPEC) as a hydrophobic block. A series of di- and tri-block copolymers is synthesized, by using macro-initiators [poly(ethylene glycol) monomethyl ethers and poly(ethylene glycol)s] of different molar masses, while systematically varying monomer to macro-initiator ratio. The synthesized block copolymers are characterized by size exclusion chromatography, for augmentation in the molar mass, NMR is used for comparison of individual block lengths alongside purity of block copolymers, with regard to presence of homopolymers of both types. Chromatographic critical conditions of the polymer synthesized by ROP of PC, PPEC, are reported for the first time. Established chromatographic critical conditions of PEG and PPEC, successfully tracked presence of homopolymers of both types, with fair comparison of individual block lengths of the block copolymers synthesized by the same macro-initiator.



Keywords: biodegradable polymers, poly(propylene ether-carbonate), amphiphilic block copolymers, liquid chromatography at critical conditions, chromatographic critical conditions of poly(propylene ether-carbonate).

1. Introduction

Covalently joined hydrophilic and hydrophobic polymeric segments generate species having completely different properties compared to their parent materials and are termed as amphiphilic block copolymers. Amphiphilic block copolymers self-assemble in selective solvent for one of the segment to form core-shell type micellar structures. Polymeric micelles have attracted considerable attention lately due to their applications in various fields of science such as surface coating, detergents, oil recovery, nanotechnology, and drug delivery etc.^{1,2} These are nano-sized vesicles that have ability to encapsulate hydrophobic drugs in the aqueous environment of body. Sustained and continuous delivery of the drug to the targeted site till the next planned dose is one of the major challenges of modern day medical science.

Poly(ethylene oxide) (PEO) is the most widely employed polymer as hydrophilic segment for designing biodegradable amphiphilic block copolymers due to its excellent biocompatibility, water solubility and non-toxicity.^{3,4} However, the selection of hydrophobic segment from a fairly long list such as polyacrylates, polyether, polyesters, and polycarbonates etc depends upon the intended application. Presence of heteroatom in the main chain of polyesters, polyethers and polycarbonates induces ability to

degrade by natural biological mechanisms. However, the speed and mechanism of degradation of polyesters, polyethers and polycarbonates vary from each other in addition to their completely different biodegradation products. Non-acidic degradation products (carbon dioxide and water) and slow degradation rate of polycarbonates compared to polyesters offers an exciting opportunity for designing novel, nontoxic, and controlled drug delivery systems.⁵⁻⁷ These degradation characteristics minimize the possibility of introduction of extra toxicity in the biological system and ensure sustained delivery of the drug to the targeted site. Amphiphilic block copolymers based on polycarbonates have been synthesized by ring-opening polymerization (ROP) of six-membered cyclic carbonates mainly trimethylene carbonate.^{8,9} However, ROP of five-membered cyclic carbonates for synthesis of amphiphilic block copolymers is not given much consideration in literature. Recently, we reported first time synthesis of amphiphilic block copolymers by ROP of ethylene carbonate (EC) using poly(ethylene glycol) monomethyl ethers (MeO-PEGs) and poly(ethylene glycol)s (PEGs) as macro-initiator.¹⁰

ROP of five-membered cyclic carbonates (EC and propylene carbonate (PC)) is associated with fractional decarboxylation, hence, the resulting polymer contains ether linkages alongside targeted carbonate linkages.¹¹⁻¹⁴ ROP of EC by anionic mechanism resulted in more decarboxylation compared to cationic mechanism, therefore, the polymer synthesized by cationic ROP con-

*Corresponding Author: Muhammad Imran Malik (mimran.malik@iccs.edu)

tains more carbonate linkages compared to the polymer synthesized by anionic ROP.^{11,12,15,16} On the same lines, we reported ROP of PC and comprehensive structure elucidation of the polymer by advanced NMR techniques.¹⁷ The polymer contains both ether and carbonate linkages similar to EC. However, the nature of ether and carbonate linkages obtained by ROP of PC are quite different from polymer synthesized by ROP of EC.^{11,17} Therefore, polymer is named as poly(propylene ether carbonate) (PPEC). The relative content of carbonate to ether linkages remained in the range of 35–42%.¹⁷

In this context, synthesis of amphiphilic block copolymers obtained by ROP of PC using MeO-PEG/PEG as macro-initiator (hydrophilic block) is reported in this study. The synthesized di- and tri-block copolymers are characterized by size exclusion chromatography (SEC) for molar mass and by NMR for average chemical composition. Block copolymers are heterogeneous materials owing to dispersity of several properties such as total molar mass, molar mass of individual blocks, chemical composition, and presence of homopolymers in the sample. To answer above-mentioned questions, block copolymers are subjected to chromatographic critical conditions of both blocks. The chromatographic critical conditions of the polymer synthesized by ROP of PC (PPEC) are reported for the first time.

2. HPLC of polymers

Chromatographic behaviour of polymers depends upon size-pore ratio and interaction parameter c of the repeat unit with the stationary phase. The interaction parameter c is measured in inverse length (nm^{-1}). Its value depends strongly on mobile phase composition (for a given polymer-sorbent system) and is independent of pore diameter, interstitial volume, pore volume and degree of polymerization. Based on the so called interaction parameter c , polymer chromatography can be sub-divided into three distinct modes of separation namely, SEC, interaction chromatography (IC) and liquid chromatography at critical condition (LCCC). The value of interaction parameter c is negative in SEC, positive in IC and equals to zero in case of LCCC.^{18–21}

SEC separates polymers according to their hydrodynamic volume in dilute solution. It is entropy controlled process and allows separation of polymers in order of decreasing hydrodynamic volume. The separation in SEC regime is realized before the void volume of the column within the limits of the pore volume. The hydrodynamic volume can then be correlated to the molar mass of the polymer standards used for calibration of the system. The exact correlation is only possible if polymer standards of same polymer having same architecture are used. Therefore, SEC can only provide molar mass relative to the used standards. Furthermore, discrimination of polymers with regard to chemical composition is not possible by SEC. However, SEC is a reasonable first test for any newly synthesized polymer with regard to augmentation in hydrodynamic size that is an indication of increase in the molar mass of the polymers.^{22–24}

The presence of polymer segments of different chemical nature in block copolymers makes situation gloomier. The interaction parameters c_A and c_B of individual structural units (A and B) may have different values. SEC separation mode will be opera-

tive if both c_A and c_B are negative. However, the hydrodynamic volumes of both structural units might be different that further complicates the situation. Nonetheless, SEC can be used as a first test for augmentation in the hydrodynamic size of the block copolymers.

Positive values of interaction parameters c_A and c_B of individual structural units (A and B) is of no practical use in analysis of block copolymers since it can result in mess of peaks that are impossible to interpret. However, IC can be employed for oligomeric separation of homopolymers and functional polymers. Solvent and temperature gradients can be used for analysis of complex polymers, however, the approach is based on the principle of altering the interaction strength of the repeat unit with the stationary phase during the run.^{18,19,25,26}

Another very important mode of liquid chromatography of polymers exists at the intersection of two extreme modes of LC of polymers, where the value of interaction parameter c equals zero. These conditions are termed as LCCC of that particular polymer in a given solvent/sorbent system. The establishment of LCCC results in molar mass independent elution of homopolymers of that particular type near the void volume of the column.^{18,19} The situation is particularly useful while analyzing block copolymers^{10,27–33} or functional polymers.^{30,34–36} The analysis can provide information with regard to molar mass distribution of non-critical block along with presence or absence of unwanted critical homopolymers in the sample. LCCC has been used for analysis of block copolymers where noncritical block elutes either in exclusion^{10,27–29} or interaction regime.^{30–33}

LCCC of hydrophilic block can be realized on reversed phase (RP) column in a highly polar mobile phase allowing hydrophobic block to elute in IC mode. On the same column, critical conditions of hydrophobic block can be realized in a non-polar mobile phase allowing exclusion of hydrophilic block. Similarly, critical conditions of hydrophilic block can be realized in a polar mobile phase on normal phase (NP) column allowing exclusion of hydrophobic block. LCCC of hydrophobic block may be possible on NP column in a mobile phase of low polarity allowing adsorption of hydrophilic segment.

3. Experimental

3.1. Material

Propylene carbonate (PC, Daejung), sodium stannate trihydrate (SS, Sigma Aldrich), diethylene glycol (DEG, Scharlau) were used as received. Polyethylene glycol monomethyl ethers (MeO-PEGs) and poly(ethylene glycol)s (PEGs) of different molar masses (MeO-PEG₅₅₀, MeO-PEG₇₅₀, MeO-PEG_{2K}, MeO-PEG_{5K}, PEG₄₀₀, PEG_{1K}, PEG_{2K}, and PEG_{4K}) were purchased from Sigma-Aldrich and were used after azeotropic distillation with toluene.

Chloroform (CHCl_3) (RCI Labscan), *N,N*-dimethylformamide (DMF), acetonitrile, (Fisher Scientific) and tetrahydrofuran (THF)-all HPLC grade were used without purification for liquid chromatography. Double deionized water was used for LC analysis by Milipore Progard 2 purification system. Chloroform-*d* 99.96% atom D with 0.5% silver wire as stabilizer (Merck) was used as a solvent for NMR analysis.

3.2. Polymerizations

ROP of PC was performed according to a recently reported procedure by our group.¹⁷ Homopolymers of PPEC (500 to 5000 g/mol) were synthesized and used as standards for establishment of LCCC of PPEC. In the preparation of block copolymers, calculated amounts of macro-initiator (MeO-PEG/PEG), monomer (PC), and catalyst (sodium stannate trihydrate-0.5 wt% of monomer) were introduced into an oven-dried flask. The polymerizations were carried out at 210 °C under continuous stirring in inert atmosphere. The copolymerization times were varied according to the targeted length of the PPEC block. The relative amounts of reagents used for synthesis of different di- and tri-block copolymers are given in Table 1.

3.3. Liquid chromatography

SEC analysis of homopolymers and block copolymers was performed on an Agilent SEC instrument (infinity 1200 series) comprising of an isocratic pump equipped with a manual injector, a column compartment, and a refractive index detector. A set of three PLgel columns (300×7.5 mm, 5 μm, Mixed D) with a PLgel guard column (50×7.5 mm) were placed in a column oven maintained at 30 °C. Chloroform was used as mobile phase at a flow rate of 1.0 mL/min. The injection volume of sample was 100 μL with concentration range 3.0 and 5.0 mg/mL. Agilent OpenLAB ChemStation software was used for calculation of molar mass on the basis of calibration curve of polystyrene (ranging from 165 g/mol to 3×10⁶ g/mol) standards procured from Polymer Standard Services (Germany).

LC/MS analysis was performed on an Agilent technologies chromatograph (infinity 1200 series) comprising of an auto-sampler, a binary pump, a column oven, and an evaporative light scattering detector (**ELSD**). The temperature of column oven was maintained at **30 °C** and mobile phase flow rate was set at **0.5 mL/min**. The chromatograms were recorded by an ELSD operating at 40 °C. The nitrogen gas pressure at nebulizer of ELSD was maintained at 50 psi. The injection volume of sample solutions was **20 µL**, having concentration in range of **3.0 and 10.0 mg/mL**. Agilent Open LAB ChemStation software was used for data analysis.

LC/MS of PPEC was performed by using above HPLC instrument with **ZORBAX ODS, 70 Å pore, 5 µm particle, 250×4.6 mm dimensions (Phenomenex, USA)**. The mobile phase for RP analysis was a mixture of acetonitrile (ACN) and double deionized water with 1.0% trimethylamine (TEA) as an additive.

LCCC of **PEG** was performed by using the HPLC instrument elaborated above with **PhenoSphere-Next 5u Sil** column, **120 Å** pore, **5 µm** particle, **250×4.6 mm** dimensions (**Phenomenex**, USA). The mobile phase used for NP analysis was a mixture of **THF** and **DMF**.

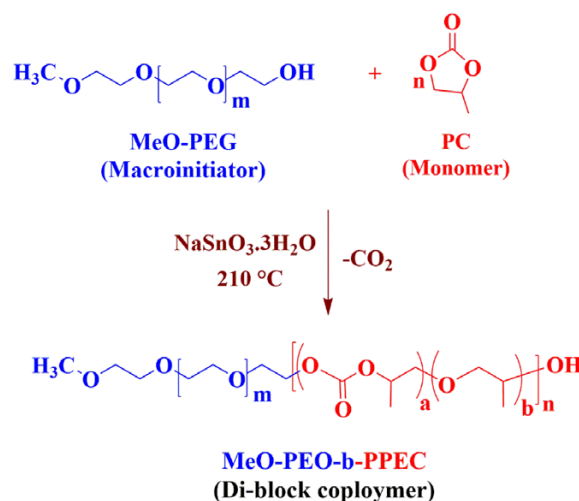
3.4. NMR analysis

NMR analyses of the block copolymers were performed on a Bruker AVANCE AN-500 MHz spectrometer. Tetramethylsilane (TMS)-d and CDCl_3 were used as reference and solvent, respectively.

4. Results and discussion

ROP of five-membered cyclic carbonates (EC & PC) is associated with decarboxylation during polymerization that leads to presence of ether linkage alongside targeted carbonate linkages in the synthesized polymers.¹¹⁻¹⁷ In our recent publication, we reported ROP of PC using sodium stannate trihydrate as a catalyst and concluded by comprehensive structural analysis by advanced NMR techniques that the polymer contains about 35-42% of carbonate linkages. The polymer obtained by ROP of PC contains carbonate linkages (equivalent to propylene carbonate) and ether linkages (equivalent to propylene oxide), hence named as poly(propylene ether carbonate), PPEC.¹⁷ Both repeating units produced by ROP of PC are hydrophobic in nature and therefore block copolymers of polyethylene glycol with PPEC should be amphiphilic in nature. Peculiar slow degradation rate and non-acid degradation products of PPEC makes them fascinating candidates for drug delivery applications. In this context, various di- and tri-block copolymers are synthesized by systematically varying the monomer to initiator ratio in the feed while using PEGs and MeO-PEGs of different molar masses as macro-initiator, Scheme 1.

Rational propagation in the molar mass of the synthesized polymer is the first question that requires attention while synthesizing any new polymer. SEC is the robust and most widely



Scheme 1. Synthesis of amphiphilic di-block copolymer (MeO-PEO-b-PPEC) by ROP of PC.

Table 1. Synthesis protocols of di- and tri-block copolymers

Sr. No.	Product Code	Macro-Initiator: Monomer (by weight)	Macro-Initiator (PEG/MeO-PEG) (g)	Catalyst (Sodium Stannate Trihydrate) (g)	Monomer (Propylene Carbonate) (g)
1	3-1	3:1	3.0	0.005	1.0
2	1-1	1:1	2.0	0.01	2.0
3	1-3	1:3	1.0	0.015	3.0

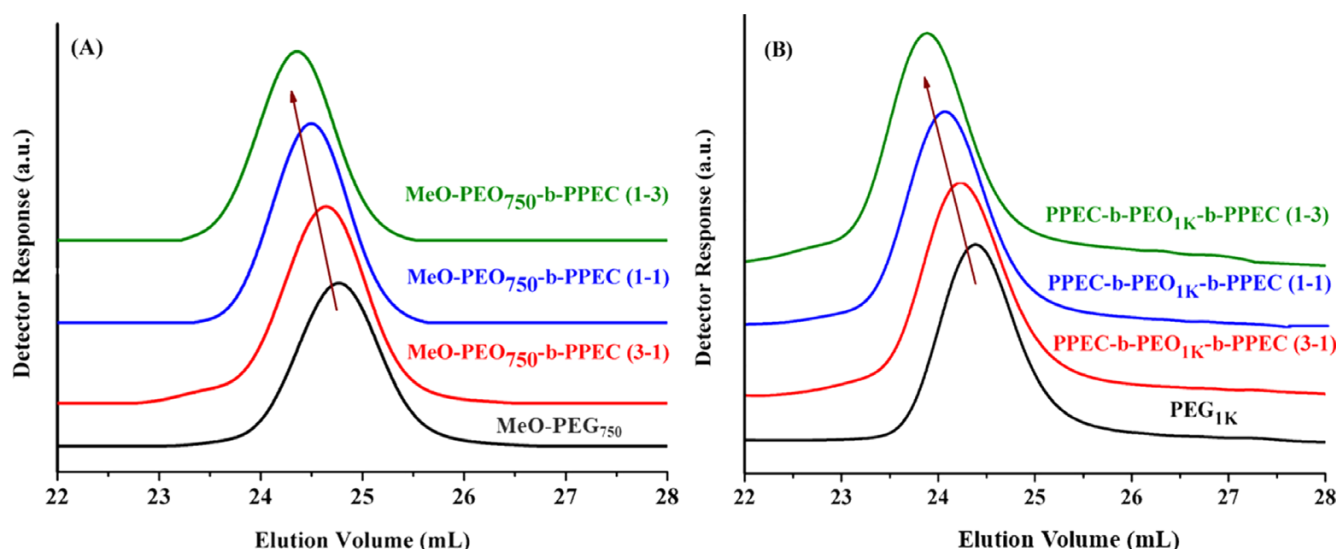


Figure 1. SEC chromatograms of (A) di-block copolymers (MeO-PEO₇₅₀-b-PPEC); (B) Tri-block copolymers (PPEC-b-PEO_{1K}-b-PPEC) synthesized using different monomer to macro-initiator ratio, on a set of three PLgel columns (300×7.5 mm, 5 μm, Mixed D) in chloroform (HPLC grade) at a flow rate of 1.0 mL/min with RI detector.

employed technique to answer this basic question. SEC separates with regard to the hydrodynamic size of the analyte in dilute polymer solution that can in turn be correlated to their relative molar mass. Figure 1 demonstrates SEC chromatograms of di- and tri-block copolymers synthesized by varying ratios of monomer to macro-initiator. As can be noticed, elution volume of the block copolymers decreased with increase in the monomer content in the initial feed. This is an expected behaviour and indicates the formation of block copolymers.

As a next step, synthesized polymers are subjected to NMR for chemical composition analysis. Figure 2 shows the ¹H NMR spectrum of PPEC-b-PEO_{1K}-b-PPEC. The signals at 1.11, 1.25, and 3.45–3.65 ppm are attributed to -CH₃ (6) of PO units, -CH₃ (1) of PC units, and -CH, -CH₂ (3,4,5,7) groups of PO, PC, EO units, respectively. The chemical composition of the block copolymers and monomer conversion can be calculated by using following equations:

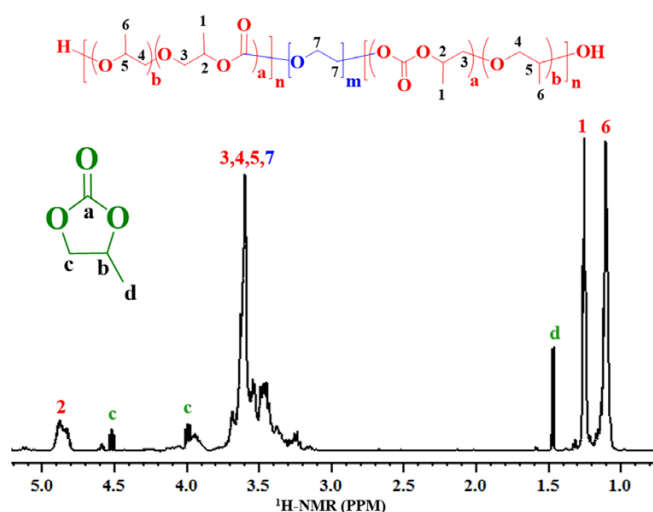


Figure 2. A representative ¹H NMR spectrum of block copolymers; PPEC-b-PEO_{1K}-b-PPEC (1:1).

$$f_{\text{PPEC}} = \frac{\left[\left(I_1 \times \frac{102}{3} \right) + \left(I_6 \times \frac{58}{3} \right) \right]}{\left[\left(I_1 \times \frac{102}{3} \right) + \left(I_6 \times \frac{58}{3} \right) \right] + \left[\left(\frac{I_{3,4,5,7}}{9} \right) - \left(\frac{I_1}{3} \right) + \left(\frac{I_6}{3} \right) \right] \times 44} \quad (1)$$

$$\text{Monomer conversion (\%)} = \frac{I_1 + I_6}{I_1 + I_6 + I_d} \times 100 \quad (2)$$

where I_1 and I_6 are integral of the signals of methyl group of PO and PC units at 1.11 and 1.25 ppm in the PPEC block, respectively. The signals at 3.45–3.65 ppm are attributed to PC, PO and EO units of PPEC and PEO. The numerical values 102, 58 and 44 correspond to the molecular weight of PC, PO and EO, respectively. Relative content of PEO as obtained by NMR varies systematically with varying monomer to initiator ratio as per expectations. The PPEC content of the products shown above (Figure 1(B)) as obtained by NMR were 23, 51 and 78%, respectively. The monomer conversion can be calculated by using equation 2, where I_1 , I_6 and I_d are integral of the signals of methyl group of PO repeat unit, PC repeat unit and propylene carbonate (monomer), respectively.

SEC and NMR revealed information regarding augmentation in the molar mass and average chemical composition. However, these techniques are unable to disclose any information regarding molar mass of the individual blocks along with presence or absence of any unwanted homopolymers in the samples. Presence of homopolymers in block copolymer samples can drastically reduce their efficiency especially in drug delivery and related applications. LCCC is the technique of choice to answer above-mentioned questions. This technique works on the principle of “chromatographic invisibility” of one of the block rendering separation with regard to other block either in exclusion or interaction regime. Establishment of critical conditions for any polymer is a thoroughly pondered, well-planned, and tedious process.^{18,19,21,37} A series of homopolymers of varying molar masses is required to realize the LCCC of any polymer. LCCC of polycarbonates have

never been reported due to commercial unavailability of polymer standards of varying molar masses. Furthermore, polymer obtained by ROP of five-membered cyclic carbonates contains ether linkages alongside carbonate linkages that further complicates the situation. Nonetheless, triad sequence of ether and carbonate linkages in PEEC and PPEC remained fairly similar throughout the polymer chain.^{11,17} This renders to similar ratio of both linkages making a pseudo homopolymer with a longer repeat unit (triad) and allowed for establishment of LCCC of PEEC.¹⁰ On the same lines, ROP of PC is also associated with fractional decarboxylation resulting in a polymer chain containing both ether and carbonate linkages, nonetheless, the triad sequence remained fairly similar throughout the chain. The synthesized polymer is therefore named as poly(propylene ether carbonate), PPEC.¹⁷ The other block of the copolymers under study is PEO that is intrinsically polar than the PPEC. Hence, its critical conditions should be established on a RP column for elution of polar PEO in SEC regime. In this regard, a binary mobile phase containing acetonitrile and water is utilized with 1% TEA as an additive. The addition of TEA is imperative to avoid unwanted polar interactions of free silanol groups of stationary phase with ether linkages.³² ACN is desorption promoting solvent while water promotes interaction of PPEC on RP columns. The elution behaviour of PPECs using various compositions of mobile phase is demonstrated in Figure 3. PPEC_{4K} eluted earlier than PPEC_{0.5K} while using only ACN with 1% TEA. This is a typical SEC behaviour, Figure 3(A). Elution volume of PPEC increased slightly by increasing water content of the mobile phase to 5% (ACN-H₂O-TEA (v/v/v%) = 94-5-1), however, elution order remains the same, B. Addition of more water in mobile phase further decreased the mobile phase strength and induced stronger interaction of the repeat units with the stationary phase. PPEC eluted in order of increasing

molar mass while using mobile phase composed of 90% ACN, 9% water and 1% TEA, Figure 3(C). Increase in the water content of mobile phase beyond 10% resulted in stronger interaction of PPEC with the stationary phase, Figure 3(D). Furthermore, PPEC eluted at similar elution volume irrespective of their molar mass while using mobile phase containing 92% ACN, 7% water and 1% TEA (2.8 mL), E. These are so called chromatographic critical conditions of PPEC on RP column. Successful establishment of critical conditions of the polymer further endorse the NMR results that ether and carbonate linkages are statistical rather than random. Hence, it behaves like a pseudo homopolymer with a longer effective repeat unit (triad) throughout the chain.^{38,39} Under established critical conditions for PPEC, PEGs should be excluded from the pores of the stationary phase. This is actually the case as can be seen in Figure 3(F).

In order to confirm the elution behaviour and selectivity of the established critical conditions of PPEC, blends of similar molar masses of PPECs and PEGs are analyzed. PPECs eluted at the same elution volume irrespective of their molar mass while PEGs are excluded from the stationary phase. The separation and resolution gets better with the increase in the molar mass of PEGs, Figure 4.

Di- and tri-block copolymers of PEG with PPEC should elute according to length of the PEO macro-initiator earlier than the elution volume of critical PPECs. Figure 5(A) depicts the elution behaviour of di-block copolymers of PPEC using MeO-PEG_{5K} as macro-initiator while varying macro-initiator to monomer ratio systematically. Di-block copolymers eluted at the elution volume corresponding to MeO-PEG_{5K} regardless of the length of PPEC block. B demonstrates the elution behaviour of several di- and tri-block copolymers synthesized by using MeO-PEG/PEG macro-initiator of varying molar masses with equal monomer to macro-initiator content in the initial feed. Block copolymers eluted earlier than the elution volume of critical PPEC with regard to the molar mass of MeO-PEG/PEG macro-initiator. There is no significant signal at the elution volume of critical PPEC which indicates the absence of any PEO-free-PPEC. In order to con-

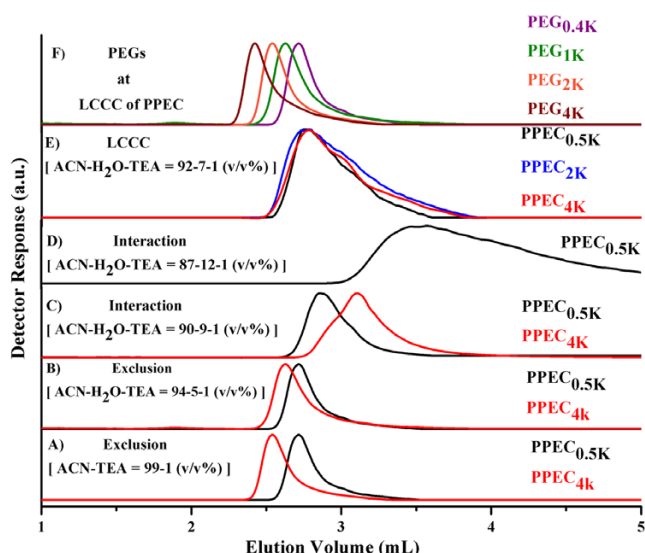


Figure 3. Elution behaviour of PPEC, Stationary phase: ZORBAX ODS RP 70 Å, 5 µm, 250×4.6 mm column, mobile phase: ACN/H₂O/TEA, detector: ELSD, column temperature: 30 °C; (A) ACN/H₂O/TEA: 99/0/1 (v/v/v%) (Exclusion); (B) ACN/H₂O/TEA: 94/5/1 (v/v/v%) (Exclusion); (C) ACN/H₂O/TEA: 90/9/1 (v/v/v%) (Interaction); (D) ACN/H₂O/TEA: 87/12/1 (v/v/v%) (Interaction); (E) ACN/H₂O/TEA: 92/7/1 (v/v/v%) (LCCC); (F) Elution behaviour of PEGs at LCCC of PPEC [ACN/H₂O/TEA: 92/7/1 (v/v/v%)].

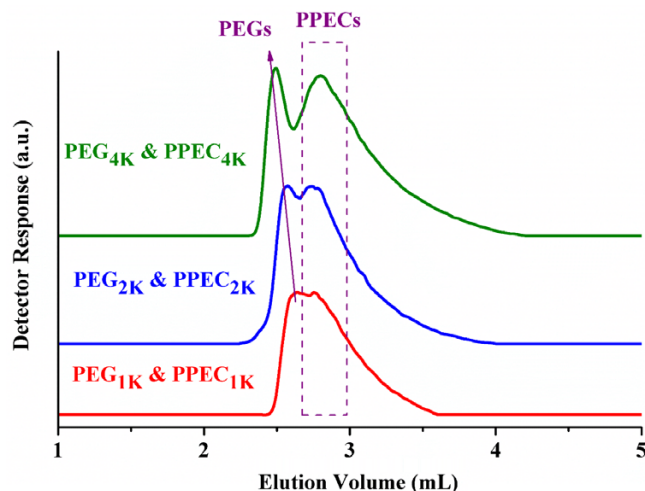


Figure 4. Chromatograms of blends of similar molar masses of PPEC and PEG at chromatographic critical conditions of PPEC on a ZORBAX ODS RP 70 Å, 5 µm, 250×4.6 mm column, detector: ELS, column temperature: 30 °C; mobile phase ACN/H₂O/TEA: 92/7/1 (v/v/v%).

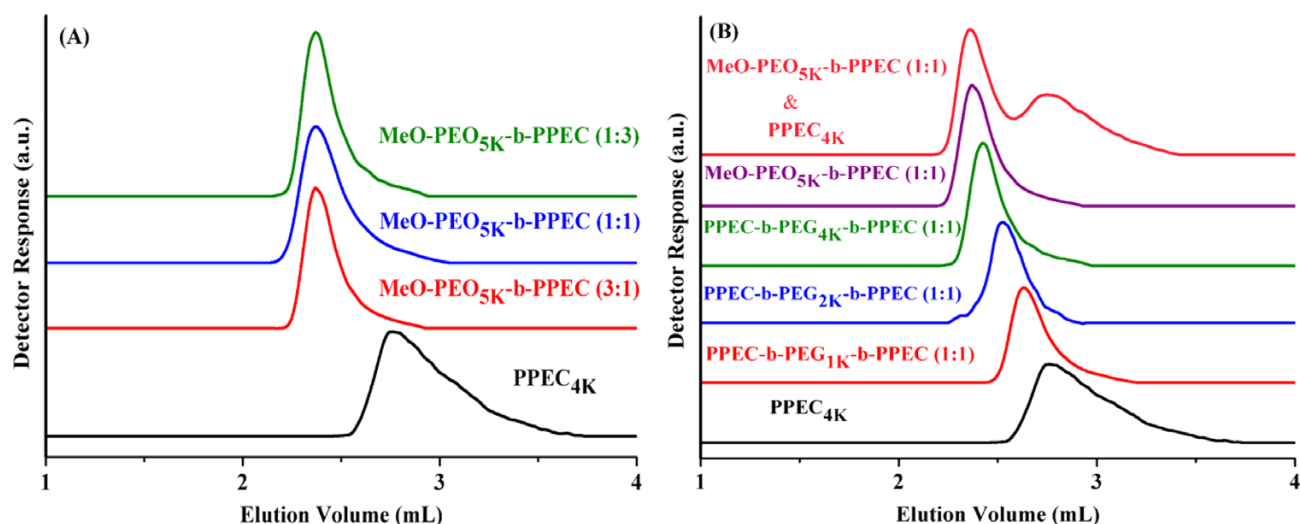


Figure 5. Chromatograms of (A) di-block copolymers (MeO-PEO_{5K}-b-PPEC) while varying initial feed ratios; (B) amphiphilic di- and tri-block copolymers of PPEC while using MeO-PEG/PEG of varying molar masses as macro-initiator and keeping equal initial feed ratio at chromatographic critical conditions of PPEC on a ZORBAX ODS RP 70 Å, 5 µm, 250×4.6 mm column, detector: ELS, column temperature: 30 °C; mobile phase ACN/H₂O/TEA: 92/7/1 (v/v/v%).

firm the assumption, one of the block copolymer is spiked with PPEC and subjected to the analysis. PPECs are clearly separated from the block copolymer at its critical conditions. The results confirm that there is no appreciable amount of PEO-free-PPECs present in the block copolymer samples.

Critical conditions of PPEC revealed information with regard to the length of non-critical PEO block and presence/absence of PEO-free-PPEC in the block copolymer samples. However, analysis at critical conditions of PEG is required for information with regard to individual block length of PPEC and presence of free MeO-PEG/PEG macro-initiator in the sample. There are several literature reports of critical conditions of PEG using different mobile phase systems on stationary phases of different polarity.^{40–42} In this case, critical conditions of PEG are required on NP column that should render exclusion of PPECs from the pores of the stationary phase. DMF is a strong desorption promoting solvent in NP chromatography where as THF promotes interaction of PEGs. The molar mass of block copolymers under study are rather low, hence, a narrow pore column (120Å) is used, Figure 6. It is pertinent to mention here that PEGs are very strongly adsorbed on the stationary phase in 100% THF. DMF strongly promotes desorption of PEG and addition of only 0.5% DMF in mobile phase resulted in elution of PEG_{0.4K} as a rather board peak, Figure 6(A). Further increase in DMF content to 1.0% promoted earlier elution of PEGs but still in interaction regime (lower molar mass eluted earlier than higher molar mass), Figure 6(B). The elution order switched from interaction to exclusion in a mobile phase composition THF/DMF: 98.0/2.0 (v/v%), Figure 6(C). It must be noticed that elution order of PEGs is exclusion, however, the elution volumes are higher than expected. This behaviour is attributed to very strong interaction of hydroxyl end-group with the stationary phase. The effect seems to be more pronounced in case of lower molar mass PEG. Stronger mobile phase composition (THF/DMF: 95.0/5.0 (v/v%)) suppresses end-group effect and allows exclusion of PEGs in the expected elution range comparable to excluded

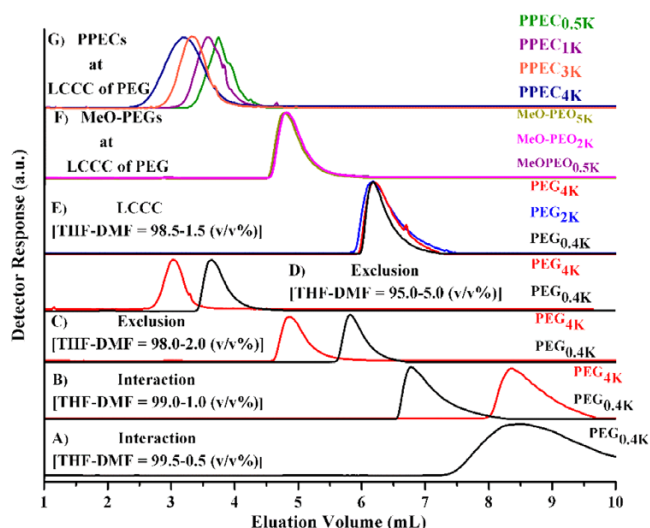


Figure 6. Elution behaviour of PEGs of varying molar masses at PhenoSphere-Next 5u Sil NP 120 Å, 5 µm, 250×4.6 mm column in mobile phase composed of THF and DMF, detector: ELSD, column temperature: 30 °C; (A) THF/DMF: 99.5/0.5 (v/v%) (Interaction); (B) THF/DMF: 99.0/1.0 (v/v%) (Interaction); (C) THF/DMF: 98.0/2.0 (v/v%) (Exclusion); (D) THF/DMF: 95.0/5.0 (v/v%) (Exclusion); (E) THF/DMF: 98.5/1.5 (v/v%) (LCCC); (F) Elution behaviour of MeO-PEGs of varying molar masses at LCCC of PEGs [THF/DMF: 98.5/1.5 (v/v%)]; (G) Elution behaviour of PPECs at LCCC of PEGs [THF/DMF: 98.5/1.5 (v/v%)].

PPECs, Figure 6(D). In between these two extreme situations at a mobile phase composition of THF/DMF: 98.5/1.5 (v/v%), all PEG standards elute at the same volume irrespective of their molar mass, Figure 6(E). These are so called LCCC of PEGs in a given solvent/sorbent system. It is pertinent to mention here that the elution volume of critical PEGs is ahead of expected void volume of the column. The late elution of PEGs at their critical conditions can be attributed to the presence of polar hydroxyl end-groups that strongly interact with the stationary phase. The assumption is further supported by elution behaviour of

MeO-PEGs of varying molar masses eluted earlier than PEGs independent of molar mass, Figure 6(F). The elution volume of MeO-PEGs is still more than expected void volume of the column due to presence of one hydroxyl end-group. Nonetheless, it confirms the strong interaction of the hydroxyl end-group with the stationary phase at critical conditions of repeat unit. Under established critical conditions of PEGs, PPECs eluted in order of decreasing molar mass earlier than the void volume of the column in a typical size exclusion mode, Figure 6(G).

As a next step, blends of similar molar masses of PEGs and PPECs are analysed at established critical conditions of PEGs. PEGs eluted at the same elution volume independent of their molar mass, however, PPECs eluted in a typical size exclusion mode earlier than PEGs, Figure 7.

Amphiphilic di- and tri- blocks copolymers should follow the elution behaviour of non-critical block (PPEC) under estab-

lished critical conditions of PEG and should elute in SEC regime in order of decreasing length of PPEC block while separating the block copolymers from the residual MeO-PEG/PEG macro-initiator if present in the sample. This is actually the case as can be seen in Figure 8. Figure 8(A) depicts the di-block copolymers prepared by using MeO-PEG₇₅₀ as macro-initiator while Figure 8(B) presents the tri-block copolymers synthesized by using PEG_{1K} as macro-initiator. The initial macro-initiator to monomer ratios are varied systematically. It is worth mentioning at this point that critical PEO block affected the elution behaviour of the block copolymers.⁴³⁻⁴⁵ Nonetheless, the block copolymers eluted earlier than the critical homopolymers that allows detection and quantification of PPEC-free-PEO in the sample. The products with lower initial monomer content have considerable amounts of unreacted macro-initiator in the samples. All MeO-PEG/PEG macro-initiator reacted in case of higher relative monomer content in the initial feed. The elution of both di- and tri-block copolymers is affected by attached critical PEO block having more pronounced effect for tri-block copolymers, Figure 8. Although, the elution volumes of block copolymers are higher than the corresponding homopolymers, the elution order follows the expected behaviour rendering earlier elution of products with higher initial monomer content compared to the products with lower initial monomer content. Off course, the estimation of non-critical block length is not possible in this case. Nonetheless, products prepared by same macro-initiator while altering the relative macro-initiator to monomer content can be compared.

The study presents first time synthesis of novel amphiphilic di- and tri-block copolymers containing PEO and PPEC. The employed analysis techniques allowed for detection of unwanted homopolymers and comparative individual block lengths of both blocks. The amphiphilic behaviour of the synthesized novel block copolymers, effect of lengths of both blocks and their relative amount on the critical micelle concentration, the size and shape of the micelles along with their drug delivery applications will be subject of our next publications of this series.

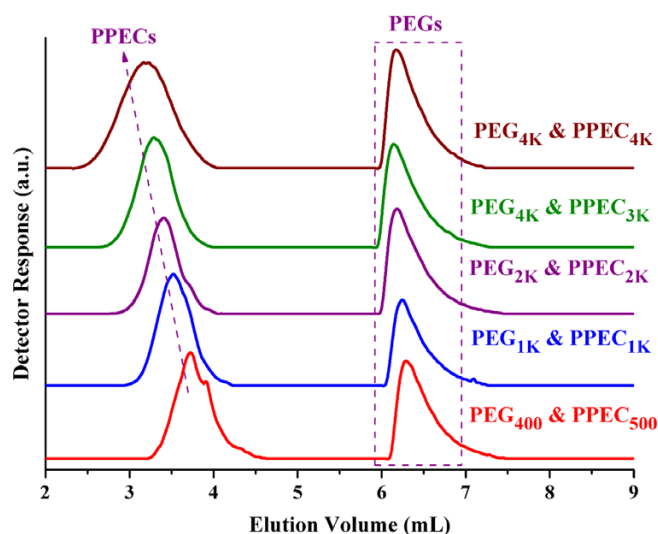


Figure 7. Blends of PEG and PPEC of similar molar masses at chromatographic critical conditions of PEG, Stationary phase: PhenoSphere-Next 5u Sil NP 120 Å, 5 µm, 250×4.6 mm column, mobile phase: THF/DMF: 98.7/1.3 (v/v%), detector: ELSD, column temperature: 30 °C.

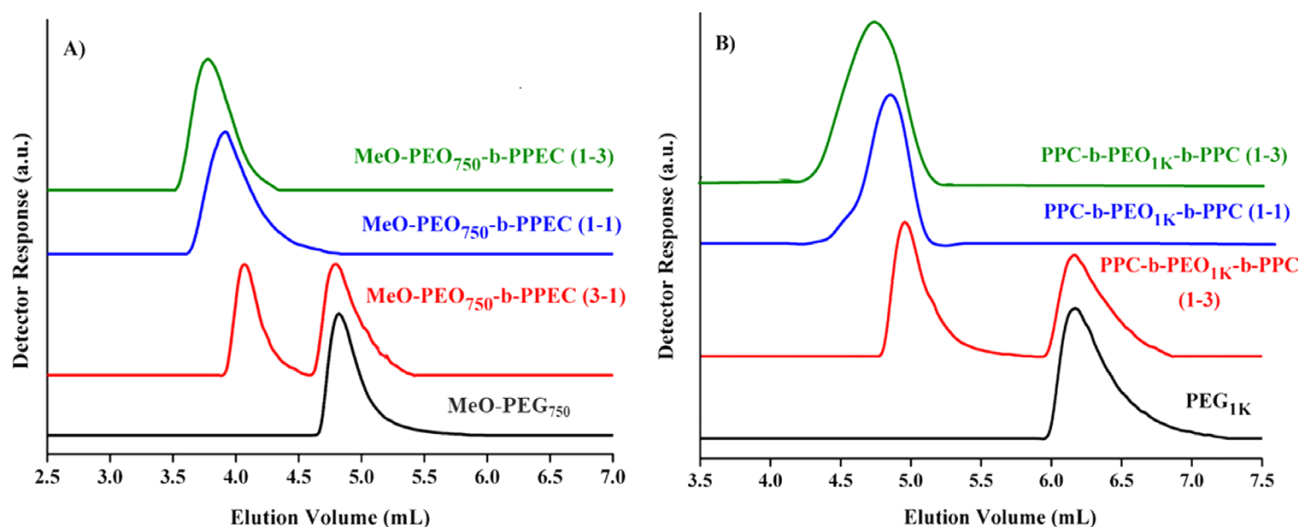


Figure 8. Chromatograms of (A) MeO-PEO₇₅₀-b-PPEC; (B) PPEC-b-PEG_{1K}-b-PPEC with varying ratios of macro-initiator and monomer at chromatographic critical conditions of PEG, Stationary phase: PhenoSphere-Next 5u Sil NP 120 Å, 5 µm, 250×4.6 mm column, mobile phase: THF/DMF: 98.7/1.3 (v/v%), detector: ELSD, column temperature: 30 °C.

5. Conclusions

Amphiphilic di- and tri-block copolymers containing PPEC as a hydrophobic block and PEO as hydrophilic part are synthesized for the first time. The synthesized novel amphiphilic block copolymers are analyzed by SEC, NMR and LCCC. LCCC of the polymer synthesized by ROP of PC, PPEC, are reported. The statistical nature of ether and carbonate units in the hydrophobic segment as revealed by NMR analysis is endorsed by chromatography. Block copolymer analysis at critical conditions allows for determination of free PEO and/or PPEC in the samples. Interactive block at critical conditions of PEG have major effect on the elution of block copolymer compared to homopolymers of excluded block. Furthermore, the products with lower initial monomer content have considerable amounts of unreacted macro-initiator in the samples. However, all macro-initiator reacted while using higher content of PC in the reaction mixture, making a part of block copolymer. The meticulous analysis protocols confirmed the successful synthesis of the block copolymers and allowed for comparison of relative length of both blocks. The synthesized biodegradable block copolymers can open new possibilities for modern drug delivery systems due to slow degradation rate and non-acid biodegradation products of polycarbonates.

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